Chapter 42 Review

Campbell Biology, AP Edition, 12e

Chapter 42 Review

Summary of Key Concepts

Homeostatic mechanisms for gas exchange reflect evolutionary strategies that have evolved in response to environmental conditions. How is this reflected in adaptations in animal systems? **(BIG IDEAS 2 & 4)**

Concept 42.1: Circulatory systems link exchange surfaces with cells throughout the body

- In animals with simple body plans, a **gastrovascular cavity** mediates exchange between the environment and cells that can be reached by **diffusion**. Because diffusion is slow over long distances, most complex animals have a circulatory system that moves fluid between cells and the organs that carry out exchange with the environment. Arthropods and most molluscs have an **open circulatory system**, in which **hemolymph** bathes organs directly. Vertebrates have a **closed circulatory system**, in which **blood** circulates in a closed network of pumps and vessels.
- The closed circulatory system of vertebrates consists of blood, blood vessels, and a two- to four-chambered **heart**. Blood pumped by a heart **ventricle** passes to **arteries** and then to the ****capillaries**, sites of chemical exchange between blood and interstitial fluid. **Veins** return blood from capillaries to an **atrium**, which passes blood to a ventricle. Fishes, rays, and sharks have a single pump in their circulation. Air-breathing vertebrates have two pumps combined in a single heart. Variations in ventricle number and separation reflect adaptations to different environments and metabolic needs.

How does the flow of a fluid in a closed circulatory system differ from the movement of molecules between cells and their environment with regard to distance traveled, direction traveled, and driving force?

Concept 42.2: Coordinated cycles of heart contraction drive double circulation in mammals

- The right ventricle pumps blood to the lungs, where it loads O2 and unloads CO2. Oxygen-rich blood from the lungs entering the heart at the left atrium is pumped to the body tissues by the left ventricle. Blood then returns to the heart through the right atrium.

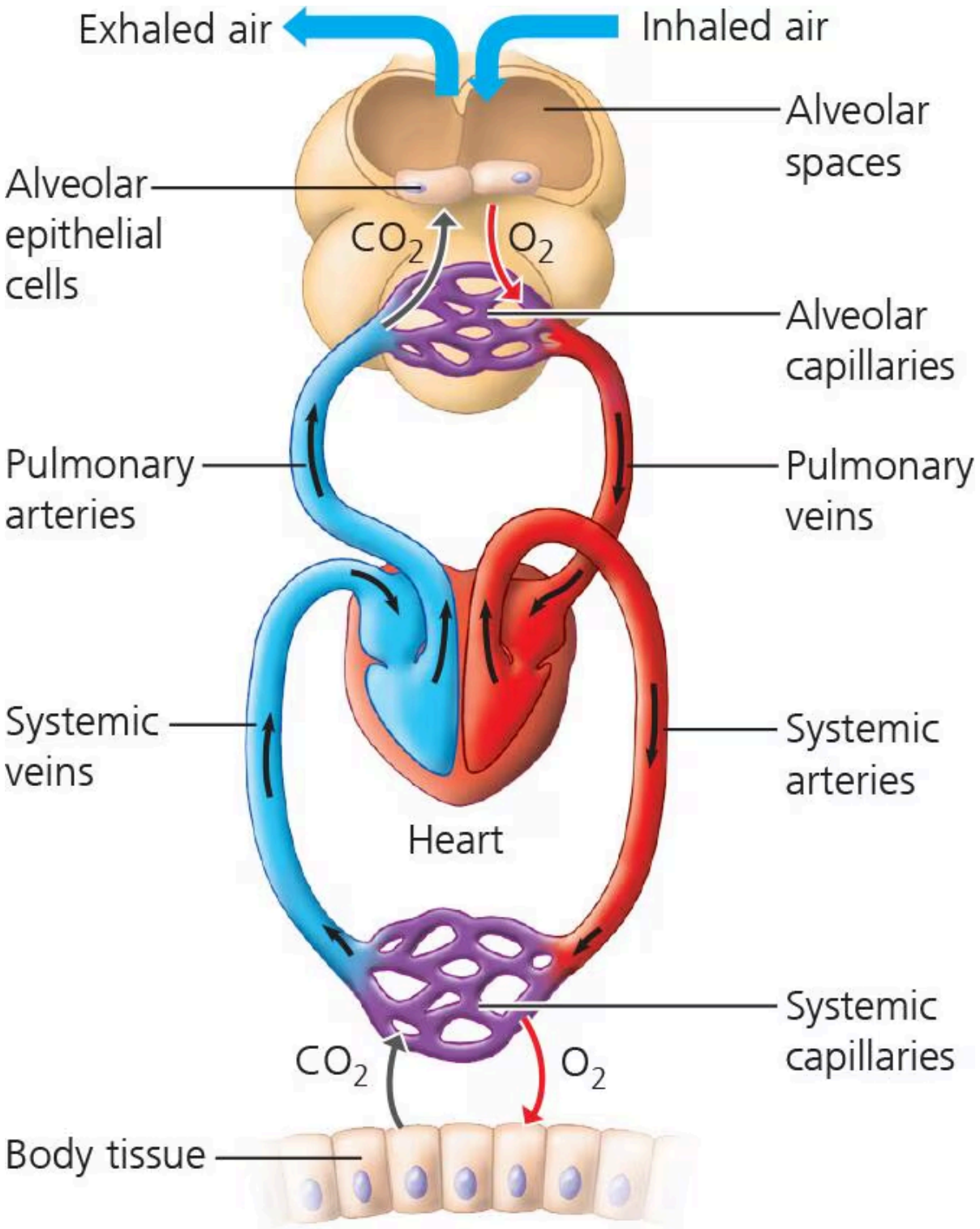

Diagram depicts path of blood through the pulmonary and systemic circuits, highlighting oxygen-rich and oxygen-poor blood and the major blood vessels. Starting in the right atrium of the heart, the path is as follows.

Oxygen-poor blood. In the heart, blood moves from the right atrium to the right ventricle and is then pumped out via the pulmonary artery to the alveolar capillaries in the lungs. As blood passes along the alveolar capillaries it takes up oxygen from inhaled air and becomes oxygen-rich. CO_2 is also unloaded and exhaled.

Oxygen-rich blood. From the alveolar capillaries, blood returns to the heart via the pulmonary veins and enters the left atrium. It then travels to the left ventricle and is pumped out via the systemic arteries to the systemic capillaries. As blood passes along the systemic capillaries, body tissue cells take up O_2 from the blood and CO_2 passes from the tissue cells to the blood. The oxygen-rich blood becomes oxygen-poor as it travels across the systemic capillary beds.

Oxygen-poor blood. Oxygen poor blood returns to the heart from the systemic capillaries via that systemic veins. It enters the heart at the right atrium to begin the double circuit again.

- During each **cardiac cycle**, a complete sequence of the heart's pumping and filling, there is a period of ventricular contraction, called **systole**, and a period of ventricular relaxation, called **diastole**. Heart function can be assessed by measuring the **pulse** (number of times the heart beats each minute) and **cardiac output**(volume of blood pumped by each ventricle per minute).
- The heartbeat originates with impulses at the **sinoatrial node** (**SA)** (pacemaker) of the right atrium. They trigger atrial contraction, are delayed at the **atrioventricular node** (**AV)**, and are then conducted along the bundle branches and Purkinje fibers, triggering ventricular contraction. The nervous system, hormones, and body temperature affect pacemaker activity.

What changes in cardiac function might you expect after surgical replacement of a defective heart valve?

Concept 42.3: Patterns of blood pressure and flow reflect the structure and arrangement of blood vessels

- Blood vessels have structures well adapted to function. Capillaries have narrow diameters and thin walls that facilitate exchange. Arteries contain thick elastic walls that maintain blood pressure. Veins contain one-way valves that contribute to the return of blood to the heart.
- The velocity of blood flow is lowest in the capillary beds as a result of their large total cross-sectional area. Blood pressure is altered by changes in cardiac output and by variable constriction of arterioles.
- Fluid leaks out of capillaries and is returned to blood by the **lymphatic system**, which also defends against infection.

If while standing you placed your forearm on your head, how, if at all, would the blood pressure in that arm change? Explain.

Concept 42.4: Blood components function in exchange, transport, and defense

- Whole blood consists of cells and cell fragments (**platelets**) suspended in a liquid matrix called ****__plasma__. Plasma proteins influence blood pH, osmotic pressure, and viscosity, and they function in lipid transport, immunity (antibodies), and blood clotting (fibrinogen). Red blood cells, or ****__erythrocytes__, transport O_2. Five types of white blood cells, or ****__leukocytes__, function in defense against microorganisms and foreign substances in the blood. Platelets function in blood clotting, a cascade of reactions that converts plasma fibrinogen to fibrin.
- A variety of diseases impair function of the circulatory system. In **sickle-cell disease**, an aberrant form of **hemoglobin** disrupts erythrocyte shape and function, leading to blockage of small blood vessels and a decrease in the oxygen-carrying capacity of the blood. In cardiovascular disease, inflammation of the arterial lining enhances deposition of lipids and cells, resulting in the potential for life-threatening damage to the heart or brain.

In the absence of infection, what percentage of cells in human blood are leukocytes?

Concept 42.5: Gas exchange occurs across specialized respiratory surfaces

- At all sites of **gas exchange**, a gas undergoes net diffusion from where its **partial pressure** is higher to where it is lower. Air is more conducive to gas exchange than water because air has a higher O_2 content, lower density, and lower viscosity.
- The structure and organization of respiratory surfaces differ among animal species. Gills are outfoldings of the body surface specialized for gas exchange in water. The effectiveness of gas exchange in some gills, including those of fishes, is increased by **ventilation** and ****__countercurrent exchange__ between blood and water. Gas exchange in insects relies on a ****__tracheal system__, a branched network of tubes that bring O_2 directly to cells. Spiders, land snails, and most terrestrial vertebrates have internal **lungs**. In mammals, air inhaled through the nostrils passes through the pharynx into the **trachea**, **bronchi**, **bronchioles**, and dead-end **alveoli**, where gas exchange occurs.

Why does altitude have almost no effect on an animal's ability to rid itself of CO_2 *through gas exchange?*

Concept 42.6: Breathing ventilates the lungs

- ****__Breathing__ mechanisms vary substantially among vertebrates. An amphibian ventilates its lungs by **positive pressure breathing**, which forces air down the trachea. Birds use a system of air sacs as bellows to keep air flowing through the lungs in one direction only, preventing the mixing of incoming and outgoing air. Mammals ventilate their lungs by ****__negative pressure__

breathing, which pulls air into the lungs when the rib muscles and ****diaphragm contract. Incoming and outgoing air mix, decreasing the efficiency of ventilation.

- Sensors detect the pH of cerebrospinal fluid (reflecting CO_2 concentration in the blood), and a control center in the brain adjusts breathing rate and depth accordingly. Additional input to the control center is provided by sensors in the aorta and carotid arteries that monitor blood levels of O_2 as well as CO_2 (via blood pH).

How does air in the lungs differ from the fresh air that enters the body during inspiration?

Concept 42.7: Adaptations for gas exchange include pigments that bind and transport gases

- In the lungs, gradients of partial pressure favor the net diffusion of O_2 into the blood and CO_2 out of the blood. The opposite situation exists in the rest of the body. **Respiratory pigments** such as hemocyanin and hemoglobin bind O_2, greatly increasing the amount of O_2 transported by the circulatory system.
- Evolutionary adaptations enable some animals to satisfy extraordinary O_2 demands. Deep-diving mammals stockpile O_2 in blood and other tissues and deplete it slowly.

How are the roles of a respiratory pigment and an enzyme similar?